

Energy Shock and Firm-Level Decarbonisation

Did the Russia–Ukraine War Reduce GHG Intensity in European Industry?

Data Group 2

Marius Borsu Lefevre Giovanni Clemente Greta Guadalupi
Petr Hrebenar Stéphane Miquel Clément Pietro Montano
Souhail Habib Nehme Joan Olaf Subirachs Garcia

Bocconi University — Geopolitics for Business

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The context

Russia's 2022 invasion of Ukraine triggered the largest European energy supply disruption in decades, sending gas prices soaring and forcing firms to rethink their energy sourcing overnight.

- Europe had built production models around cheap Russian pipeline gas
- The shock arrived as climate commitments tightened: EU Fit for 55, CSRD
- Energy-intensive industries — chemicals, metals, cement, paper — were hit hardest
- Yet the literature on geopolitical shocks and environmental performance is thin

Did the Russia–Ukraine energy shock trigger a **reduction in firm-level GHG intensity** among European energy-intensive firms — and was this effect **different** for firms in countries more or less dependent on Russian gas?

- **Outcome:** GHG Direct Intensity — Scope 1 emissions (tCO₂e) per \$1M revenue
- Measured by **TruCost (S&P Global)**: captures *operational* emissions efficiency, normalised for firm size, with lowest measurement noise across datasets (NGFS 2024)

Finding: Gas Dependency Attenuates Decarbonisation

Model 3

$\hat{\beta}_2 = -43.5^{***}$, $\hat{\beta}_3 = +9.8^{***} \Rightarrow$ net effect (doubly treated): **-33.7 tCO₂e/\$M**

- Gas-dependent countries faced **constrained transition paths** (LNG bottlenecks, long renewable lead times)
- Standalone gas-exposure effect: +5.2 in Model 1 $\rightarrow$ indistinguishable from zero (-2.6, $p = 0.30$) in Model 3

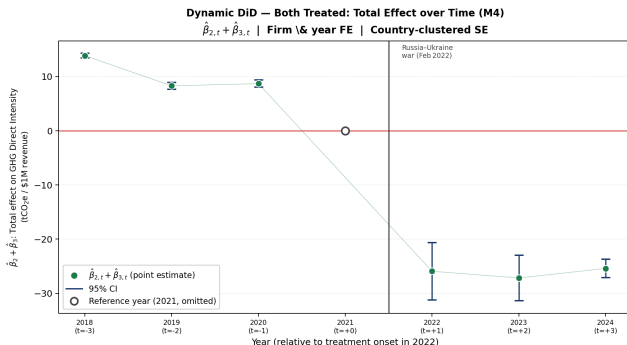


Figure: Dynamic DiD: total effect for doubly treated firms / $\hat{\beta}_2$ Ref: 2021

Main finding

The RU-UA energy shock **significantly reduced GHG intensity** among European EII firms: -38 – -43 tCO₂e/\$M (32–36% of sample mean). Effect is **smaller** in gas-dependent countries (≈ -10 tCO₂e/\$M attenuation).

Policy implications

Country level

Energy diversification (interconnectors, LNG, renewables) is both an energy-security *and* decarbonisation imperative.

Firm level

Prioritise energy-mix flexibility and long-term power-purchase agreements to decouple GHG performance from host-country endowments.